

Section A : Pure Mathematics [40 marks]

- 1** In a computer game, a player is given a plot of land to grow tomatoes, pumpkins and cherries. The player can only plant one type of fruit on the plot of land and the land can only produce 1 kg of fruits each time. The time taken from planting to harvesting the fruits, production cost and selling price for 1 kg of each type of fruit are shown in the table below.

	Time required (Hours)	Production cost (\$)	Selling price (\$)
Tomatoes	3	15	35
Pumpkins	5	k	50
Cherries	16	45	100

The total time taken for Jimmy to plant and harvest 23 kg of fruits was 154 hours. The total production cost was \$545 and he made a profit of \$735 from selling the harvested fruits. Find the value of k . [4]

[You are to assume that Jimmy harvested the fruits once they were ready and there was no time lapse between harvesting and planting new fruits.]

- 2** (i) Show that $x^2 - 2x + 2$ is always positive for all real values of x . Hence, or otherwise, solve the inequality $\frac{x}{x-4} - \frac{1}{2x-3} \geq 0$. [4]

- (ii) Hence, solve $\frac{x^2}{x^2-4} - \frac{1}{2x^2-3} < 0$. [3]

- 3** (a) Differentiate $(\ln \sqrt{2x})^3$ with respect to x . [3]

- (b) The equation of a curve is $y = \frac{a}{2x-a}$, where a is a positive constant.

The point A on the curve has coordinates $(a, 1)$.

- (i) Find the equation of the tangent at A . [2]
 (ii) If the tangent to the curve at A meets the x -axis at P and the y -axis at Q , and the length of PQ is 7 units, find the exact value of a . [3]

4 (a) Find the exact value of k such that $\int_1^2 \frac{1}{2x-1} dx = \int_1^5 \frac{k}{\sqrt{2x-1}} dx$. [3]

(b) (i) On a single diagram, sketch the graphs of $y = 4e^{-x^2} - 1$ and $y = \frac{2x+1}{1-x}$,

stating clearly the equations of any asymptotes and the coordinates of any points of intersection with the y -axis. [3]

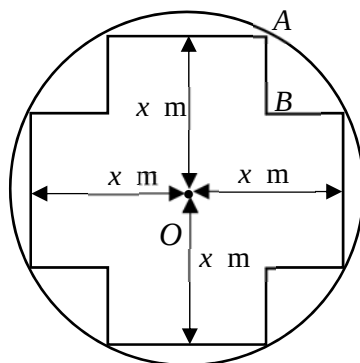
(ii) Find the x -coordinate of the point of intersection of $y = 4e^{-x^2} - 1$ and $y = \frac{2x+1}{1-x}$, giving your answer correct to 5 decimal places. [1]

(iii) Write down as an integral an expression for the area of the region bounded by $y = 4e^{-x^2} - 1$, $y = \frac{2x+1}{1-x}$ and the y -axis. Evaluate this integral, giving your answer correct to 4 decimal places. [2]

5 (a) By expressing $x\sqrt{r^2 - x^2}$ where r is a constant, in the form of $\sqrt{f(x)}$, prove that

$$\frac{d}{dx} \left(x\sqrt{r^2 - x^2} \right) = \frac{r^2 - 2x^2}{\sqrt{r^2 - x^2}}, \text{ where } 0 < x < r. \quad [3]$$

(b) The diagram shows a flower-bed in the shape of a symmetric cross inscribed in a circle of fixed radius r m with centre O .



(i) By expressing AB in terms of r and x , prove that the area of the cross, A m², is given by

$$A = 8x\sqrt{r^2 - x^2} - 4r^2 + 4x^2 \quad [3]$$

(ii) Prove by differentiation, that the value of x for which the area A is maximum will satisfy the equation $5x^4 - 5r^2x^2 + r^4 = 0$. [3]

Let the radius of the circle be 5 m for the rest of the question.

(iii) Find the value of x for which the area A is maximum. [3]

Section B : Probability and Statistics [60 marks]

- 6 A machine is used to generate codes consisting of 4 characters. A character is either one of the 26 letters of the alphabet A to Z or one of the 10 digits 0 to 9. Each of the 36 characters are equally likely to be generated and each character can be generated more than once.

Find the probability that a randomly generated code has

- (i) no digits, [1]
(ii) three different letters and one digit. [3]

- 7 A factory manufactures paperweights consisting of glass mounted on a wooden base. The volume of glass, in cm^3 , in a randomly chosen paperweight has a normal distribution with mean 56.5 and standard deviation 2.9 and the volume of wood, in cm^3 , has an independent normal distribution with mean 38.4 and standard deviation σ .

The probability that the total volume of a randomly chosen paperweight exceeds 100 cm^3 is 0.05. Find the value of σ . [4]

- 8 A shop sells mangoes packed in boxes of 12. It is given that p % of the mangoes are rotten.

- (i) Given that the mean number of rotten mangoes in a box is 2.4, show that the value of p is 20. [1]

Use $p = 20$ for the rest of the question.

- (ii) A buyer inspects a randomly chosen box of mangoes. Find the probability that

- (a) at least 2 mangoes are rotten, [1]
(b) the 10th mango inspected is the second mango that is rotten. [2]

Explain why the answer to part (b) is smaller than the answer to part (a). [1]

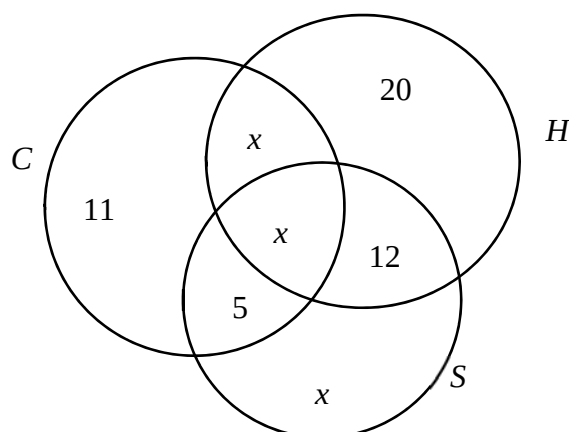
- (iii) The shop sells 60 boxes of mangoes. Find the probability that the mean number of rotten mangoes per box is less than 2. [3]

[Turn over

- 9** A car manufacturer is testing the braking distance for a new model of car. The table shows the braking distance, y metres, for different speeds, x kmh^{-1} , when the brakes are applied.

x	30	50	70	90	110	130
y	25	50	85	155	235	350

- (i) Give a sketch of the scatter diagram for the data, as shown on your calculator. [2]
- (ii) Find the product moment correlation coefficient and comment on its value in the context of the data. [2]
- (iii) Find the equation of the regression line of y on x in the form $y = ax + b$, giving the values of a and b correct to 3 significant figures. Sketch this line on your scatter diagram. [2]
- (iv) Use the equation of your regression line to calculate estimates of braking distances when the speeds of cars are 100 kmh^{-1} and 150 kmh^{-1} . Comment on the reliability of your estimates. [2]
- (v) A seventh set of data (u, v) is added and the equation of the regression line of y on x remains unchanged. State the values of u and v . [1]



A college organised Overseas Community Involvement Programme trips to Chiang Mai, Hainan and Surabaya. During the interview, a group of selected students was asked if they had ever visited Chiang Mai, Hainan and Surabaya. The number of students who had visited the 3 different places are shown in the Venn diagram. It was found that every student had visited at least one place. One student is selected at random, and the events C , H and S are defined as follows.

- C is the event that the student visited Chiang Mai.
- H is the event that the student visited Hainan.
- S is the event that the student visited Surabaya.

- (i) Find $P(C)$ and $P(C|H)$, giving your answers in terms of x . [2]
- (ii) Given that $P(C) = P(C|H)$, find the value of x . [2]
- (iii) Explain, in the context of this question, what is meant by $P(C'|S')$, and find its value. [4]
- (iv) Hence, state with a reason, whether C and S are independent. [2]

11 A glass manufacturer produces large batches of jars. The weights of jars have a standard deviation 10.5 g. The manufacturer claims that the mean weight of a jar is at least 502 g. A random sample of 50 jars is taken and the mean weight is 499 g.

- (i)** State appropriate hypotheses for the test, defining any symbols you use. [2]
- (ii)** Test, at the 1% significance level, whether the manufacturer's claim is valid. [3]
- (iii)** Explain the meaning of "at the 1% significance level" in the context of the question. [1]

A second sample of 60 jars is taken and the weights of the jars, x grams are summarised by

$$\sum (x - 502) = -222, \quad \sum (x - 502)^2 = 8401.13.$$

- (iv)** Find unbiased estimates for the population mean and variance using this second sample. [3]

A test, at the $\alpha\%$ significance level, shows that there is sufficient evidence to suggest that the population mean weight of jars from this second sample differs from 502 g.

- (v)** Find the set of possible values of α . [4]

- 12 A supermarket sells two types of strawberries, *A* and *B*. Each type of strawberry comes in different sized packaging. The masses, in kg, of a packet of type *A* strawberries and a packet of type *B* strawberries are modelled as having independent normal distributions with means and standard deviations as shown in the table.

Strawberries	Mean	Standard deviation
Type <i>A</i>	1.2	0.2
Type <i>B</i>	1.1	0.1

Type *A* strawberries are sold at \$20 per kg and type *B* strawberries at \$26 per kg.

Audrey picks one packet of type *A* strawberries and one packet of type *B* strawberries.

- (i) Find the probability that none of the two packets picked by Audrey weighs less than 1.2 kg. [2]
- (ii) The probability that the total weight of the two packets chosen by Audrey lies between 2.2 kg and m kg is 0.45, where $m > 2.2$. Find the value of m . [3]
- (iii) Find the probability that the total cost of the two packets picked by Audrey will not exceed \$50. [3]
- (iv) Jackson buys one packet of type *A* strawberries and three packets of type *B* strawberries. Find the probability that the packet of type *A* strawberries weighs more than the average weight of the three packets of type *B* strawberries by less than 0.2 kg. [4]